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Orthogonal Functionalisation of Upconverting $NaYF_4$ Nanocrystals

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Abstract: A simple and straightforward method for the orthogonal functionalisation of upconverting $NaYF_4$ nanocrystals (UCNCs)—doped with Yb^{3+} and Er^{3+}—based on *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide/*N*-hydroxysuccinimide (EDC/NHS) selective reactions between two dyes and two different reactive groups present at the periphery of the upconverting nanocrystals is reported. Organic-soluble UCNCs of 10 and 50 nm in size are encapsulated efficiently in a 1:1 mixture of two commercial 3000 Da poly(ethylene glycol) derivatives with two different reactive groups (amino and carboxylic groups). The water-dispersible UCNCs are non-cytotoxic, stable in the physiological environment, and present free amine and carboxylic reactive groups on their periphery, allowing rapid, selective, and modular covalent conjugation to payloads through EDC/NHS reactions. PEG-encapsulated UCNCs with and without covalent conjugation to payloads are characterised in vitro through spectroscopic, dynamic light scattering, and electron microscopy measurements. Living cell analyses coupled with TEM measurements confirm the uptake and low cytotoxicity of the coated UCNCs. They are linked covalently to two different dyes, internalised by living cells, and analysed by confocal microscopy. The related colocalisation measurements prove the reactivity of both amines and carboxylic acids on the periphery of the nanocrystals. This approach demonstrates that it is possible to produce water-dispersible and cyto-compatible dual-functional UCNCs.

Keywords: lanthanides · nanocrystals · nanoparticles · polymers

Introduction

$NaYF_4$ upconverting nanocrystals (UCNCs) doped with 18% Yb^{3+} and 2% Er^{3+} are triggered by low-power 980 nm laser diodes, and are able to emit light in the visible (400–600 nm) and NIR (600–800 nm) ranges through a unique multi-photon excitation process.[1] Owing to this peculiar behaviour, the application of UCNCs to nanomedicine[2] is expected to result in the development of novel nanosystems, in particular for theranostics. For the synthesis of highly luminescent and geometrically controlled UCNCs, the reactions are usually performed in high-boiling, non-coordinating solvents such as octadecene, in the presence of coordinating ligands (typically oleic acid).[1,3] The resulting UCNCs are hydrophobic, and thus, it is impossible to exploit their features directly for biological applications or to investigate their behaviour in water and physiological media. Therefore, the development of a robust and versatile post-synthesis route that provides non-toxic, stable, and hydrophilic UCNCs is mandatory for the full development of their potential in nanomedical fields.

To date, several strategies have been proposed to provide functionalisable water-dispersible UCNCs for biomedical applications.[2b,3b] Among them, the main post-synthetic encapsulation processes for UCNCs stem from the use of amphiphilic polymers,[4] layer-by-layer deposition,[5] silica or host shells,[3b,6] ligand exchange,[2a,7] and ligand oxidation.[8] Commonly, the above methods use non-commercial ligands or hide the UCNC surface with a silica shell, preventing investigations on some likely synergistic effects.[9] For the improvement of their versatility, a direct and efficient process to achieve nanostructures bearing either one reactive group or two orthogonal reactive groups ready to couple (bio)molecules of interest through covalent linkage is desirable.[10] The most attractive covalent conjugation should be highly efficient, specific, based on orthogonal chemistry approaches, able to be performed in a range of environments under relatively mild conditions, and followed by simple purification procedures, such as peptide (EDC/NHS) reac-

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tions.[11] The resulting nanocrystals should then be suitable for the production of polyvalent nanostructures that can be used in nanomedicine for applications such as multi-signalling nano-probes or theranostics nano-tools.[12]

Herein, we present an efficient and simple post-synthetic encapsulation protocol for the production of biocompatible, water-dispersible UCNCs that exhibit reactive carboxylic and amine groups on their surface coating. This approach is suitable for the production of smart UCNCs, which are stable in physiological media and readily and modularly linkable with up to two distinct selected (bio)molecules and/or fluorophores.

The encapsulation process was carried out on $NaYF_4$ nanocrystals doped with Yb^{3+} and Er^{3+}, synthesised through one of the standard methods reported in the literature,[13] using commercial hetero bifunctional PEGs with a thiol group at one end and an amine or carboxylic group at the other (HS-PEG-X, where X is NH_2 or COOH). The UCNCs were encapsulated by HS-PEG-CO_2H, HS-PEG-NH_2, or a mixture of the two different PEGs in a 1:1 molar ratio in order to obtain nanocrystals bearing one or two orthogonal reactive groups. In this protocol, the PEG ligands are used as both the capping ligand and the water-stabilising agent. PEG moieties can function as polydentate ligands and bind to lanthanide ions (Ln^{3+} ions), which are oxophilic, and some of which are located on the UCNC surface. We presumed that the thiol group could eventually become physi- or chemisorbed (the thiol–Ln binding energy has already been estimated[14]) to the UCNC surface, aided by the previous coordination of ether groups close to this group. PEG molecules are widely used as capping agents for different kinds of nanoparticles applied in biology and medicine,[15] in particular because PEGs 1) prevent undesired protein adhesion on nanostructures,[15a,16] and 2) yield non-toxic and non-immunogenic nanostructures,[15a] improving their in vivo biocompatibility.[17]

The success of the PEG-encapsulation process of different-sized UCNCs (average diameter of about 10 and 50 nm) was evaluated through spectroscopic techniques and in living cells. For demonstration of the versatility of PEG-coated UCNCs, these nanocrystals were covalently conjugated to one or two kinds of fluorophores by using peptide chemistry, and the corresponding colocalisation measurements were made in living cells.[10,11] These data show the performance of the coating proposed here and its usefulness in building new in vivo multitasking biomarkers/sensors within a single nanostructure.

Results and Discussion

Here, we report an efficient and straightforward method for yielding mono- or dual-functional, non-toxic UCNCs that are readily and covalently conjugable to a broad spectrum of biomolecules and fluorophores through simple peptide (EDC/NHS) chemistry. This protocol is applied to $NaYF_4$ nanocrystals (10 and 50 nm in size) doped with Yb^{3+} and Er^{3+}, which are synthesised through a standard method reported in the literature using a high-boiling-point solvent (octadecene).[13] Specifically, we show the results on 10 nm nanocrystals, but the encapsulation process is also applicable to larger crystals (up to 50 nm, see Figure 1 and Figure S1 of the Supporting Information).

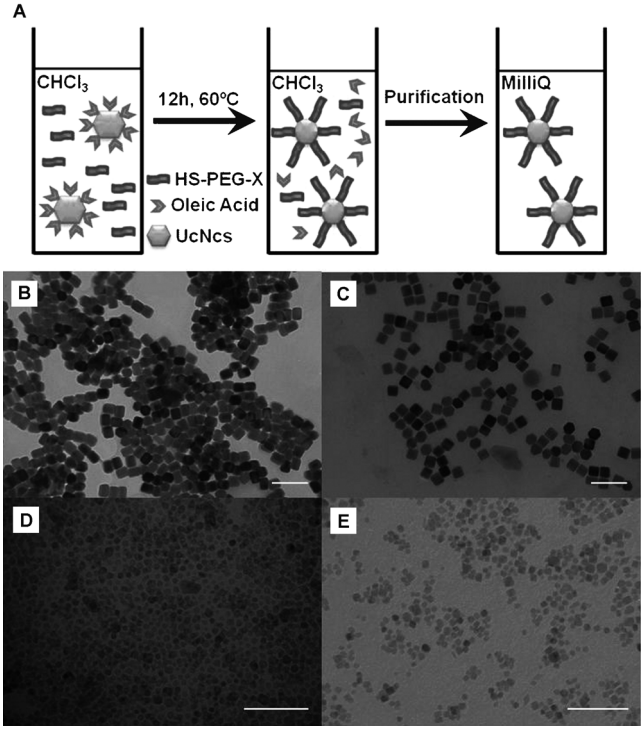

Figure 1. A) Schematic illustration of the UCNC encapsulation process. See text for a complete discussion. B)–E) TEM images of UCNCs before (B, D) and after (C, E) capping 50 nm (B, C) and 10 nm (D, E) UCNCs with Pcn. Scale bars are 200 nm for B and C and 100 nm for D and E. In this work, Pc is HS-PEG-COOH, Pn is HS-PEG-NH_2, and Pcn is a 1:1 mix of the two polymers.

Figure 1 A illustrates the encapsulation process. The reaction was performed using two different thiol-modified hetero-bifunctional PEGs with a thiol group at one end and an amine (Pn, HS-PEG-NH_2) or carboxylic (Pc, HS-PEG-COOH) group at the other to obtain nanostructures bearing one or two different reactive groups (amines and/or carboxylic acids) on their periphery. PEG is a flexible, hydrophilic, and non-toxic molecule that is approved by the Food and Drug Administration (FDA) for many medical uses in humans.[18] Furthermore, PEG chains confer a steric repulsive stabilisation of the nanoparticles, preventing agglomerations and non-specific interactions with proteins and cells.[16] The protective mechanism of PEGs on nanostructures[18] has been associated with 1) the high flexibility of the polymer chain, which enables free rotation of the polymer units, and 2) the strong hydrophilicity, which creates a stabilising shield around them. In the encapsulation process proposed here, we exploit the ability of PEG to dissolve in both organic and aqueous phases (Figure 1 A).

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Indeed, the process consists of two steps, the first being PEG encapsulation of UCNCs in chloroform, and the second being the extraction of the nanocrystal in milliQ water. In the first step, an excess[2a] of Pn or Pc or of a 1:1 mixture thereof is added to a chloroform solution of UCNCs to ensure the complete exchange of the original ligand (oleic acid). The lanthanide–thiol (Ln–S) binding[19] is a thermodynamically favoured process,[14] but when Pc was used, we had to perform the ligand exchange at 60°C to ensure the success of the lanthanide–thiol over the competitive lanthanide–carboxylate interaction.[14,20] Thus, the physi- or chemisorption of the ligand to the UCNC surface through the thiol group could eventually be facilitated by the previous coordination of the PEG ether groups that are close to the thiol group.

Thus, three kinds of water-dispersible PEGylated UCNCs were obtained by simply selecting the desired reactive group(s) at the end of the polymer: Pn (amines), Pc (carboxylic acid), and the two orthogonal reactive groups Pcn (amine and carboxylic acid by using a 1:1 mixture of Pc and Pn).

In the second step, the solutions were purified and dried, and only the fully capped UCNCs were dissolved by sonication in milliQ water (see Experimental Section). Figure 1 shows TEM images of the UCNCs before (B and D) and after (C and E) the capping process with Pcn (see Figure S1 for TEM images of UCNCPc and UCNCPn). In the case of the oleate-capped crystals, the 2D layer is highly ordered, whereas the spacing between the nanostructures increases in the PEG-encapsulated UCNCs. This suggests that the interactions between nanoparticles change because of the substitution of oleate by the polymer in both the smaller and larger nanostructures (of ≈10 and 50 nm, respectively). The TEM images of the PEG-capped UCNCs also show well-separated particles with no signs of agglomerations, indicating that the PEG ligands can stabilise the crystals effectively in water. Moreover, dynamic light scattering (DLS, data not shown) and IR spectroscopy (Figure 2) measurements are in good agreement with data reported in the literature for similar coatings,[21] and also confirm the successful derivatisation process. The hydrodynamic sizes of UCNCPc, UCNCPcn, and UCNCPn (UCNCs of 10 nm in diameter) were 64.8, 56.3, and 72.4 nm, respectively (Figure S2, Supporting Information). This gap between the TEM and DLS measurements is reasonable, because DLS analyses the Brownian motion of the solvated particles, and therefore, the measurement corresponds to the hydrodynamic size, whereas only the core diameter of the UCNCs can be measured by TEM.[22]

IR spectra (Figure 2) of oleate and ligand-exchanged UCNCs were recorded. The PEG-capped nanostructures exhibited some typical bands of the PEG ligands (Figure S3 of the Supporting Information): 1) at 1600, 1470, and approximately 1100 cm^{-1} (broad) for UCNCPc, and 2) at 3400 cm^{-1} (strong), between 900 and 1300 cm^{-1} (strong), and 650 cm^{-1} (weak) for UCNCPn. It is important to note that the ν(C=O) stretching vibration of the carboxylic acid was not detected in the UCNCPc; similar behaviour has previously been reported for gold nanoparticles decorated with carboxylic acid moieties.[23] Thus, taking into account the IR spectrum of the as-synthesised UCNCs (coated with oleate) as a "reference", the presence of these new peaks in the PEG-encapsulated samples (Figure 2) confirms the successful coating process. Interestingly, the spectrum of the UCNCPcn NPs bearing the two reactive groups shows characteristic peaks of Pc (1470 cm^{-1}) and Pn (3400 cm^{-1}, between 900 and 1300 cm^{-1}, and 650 cm^{-1}). These data are in accordance with the presence of both types of PEGs on the periphery of the UCNCs.

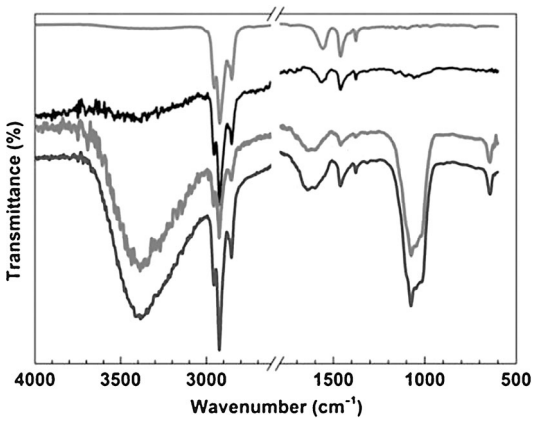

Figure 2. Top to bottom: FTIR spectra of solid samples of UCNCs coated with oleate, Pc, Pn, and 1:1 PcPn mix. For the PEG-encapsulated UCNCs there are characteristic peaks such as at 650, 3400, and between 900 and 1300 cm^{-1} for UCNCPn, and at 1600, 1470, and 1100 cm^{-1} for UCNCPc. As expected, the spectra of UCNCPcn show characteristic bands of UCNCPc and UCNCPn.

The synthesised PEG-encapsulated UCNCs were dispersed in various solvents to test their versatility. Then, their fluorescence spectra were recorded by using 980 nm laser irradiation (usually 189 mW). Figure 3 shows the emission spectra of oleate-capped UCNCs in chloroform and UCNCPc, UCNCPn, and UCNCPcn in milliQ water. Because of the high solubility of PEG-capped UCNCs in various media, their emission spectra were also measured in acetonitrile, methanol, THF, chloroform, and dichloromethane (for an example in acetonitrile, see Figure S4 in Supporting Information; other solvents not shown). The mechanism responsible for the upconversion luminescence is well known.[3c,24] The upconversion emission spectra of our NPs are consistent with others reported previously[2a,3b,c,25] for cubic and hexagonal $NaYF_4$: 2% Er^{3+}, 18% Yb^{3+}, showing the violet, green, and red upconversion emissions of Er^{3+}, which correspond to the $^2H_{9/2}\rightarrow{}^4I_{15/2}$, ($^2H_{11/2}$, $^4S_{3/2}$)$\rightarrow{}^4I_{15/2}$, and $^4F_{9/2}\rightarrow{}^4I_{15/2}$ transitions, respectively (Figure S5, Supporting Information).

Interestingly, the green to red ratio[26] did not change significantly between UCNCPn, UCNCPc, and UCNCPcn, either in milliQ water or in acetonitrile. A slight increase in the red emission of PEG-coated UCNCs was observed when compared with that of the oleate-capped NPs (Fig-

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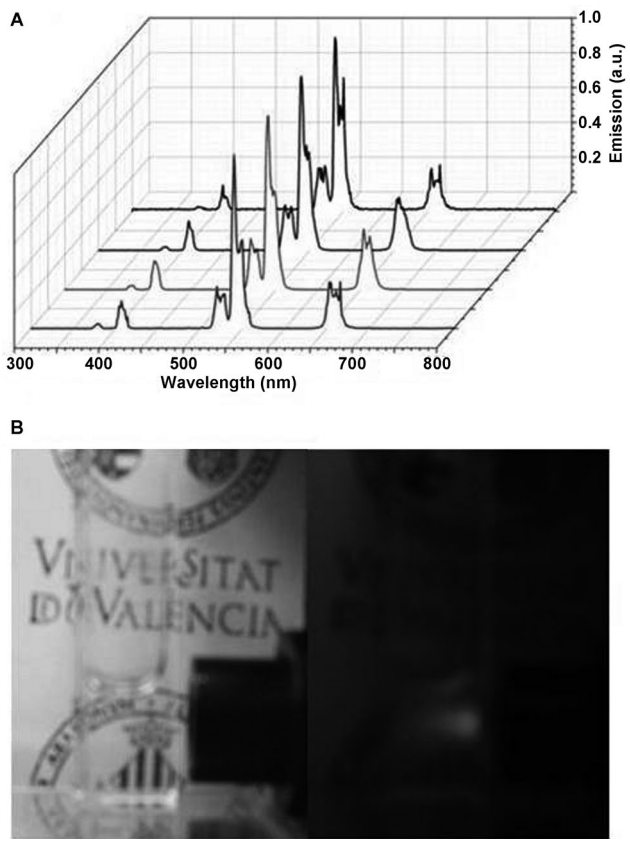

Figure 3. A) Top to bottom: emission spectra under 980 nm (189 mW) laser light irradiation of UCNCs encapsulated with oleic acid (in chloroform solution), PEG-COOH (in milliQ water), 1:1 mix (in milliQ water), and PEG-NH_2 (in milliQ water). The G/R ratio does not vary significantly (see also Figure S5 in Supporting Information). Emission spectra are normalised at 540 nm. B) Pictures of 2 mg mL^{-1} milliQ water solution of UCNCPcn irradiated with white light (left) and 980 nm laser light (right). The power of the triggering laser was 189 mW and all images were taken by activating the auto-mode collection option of the camera.

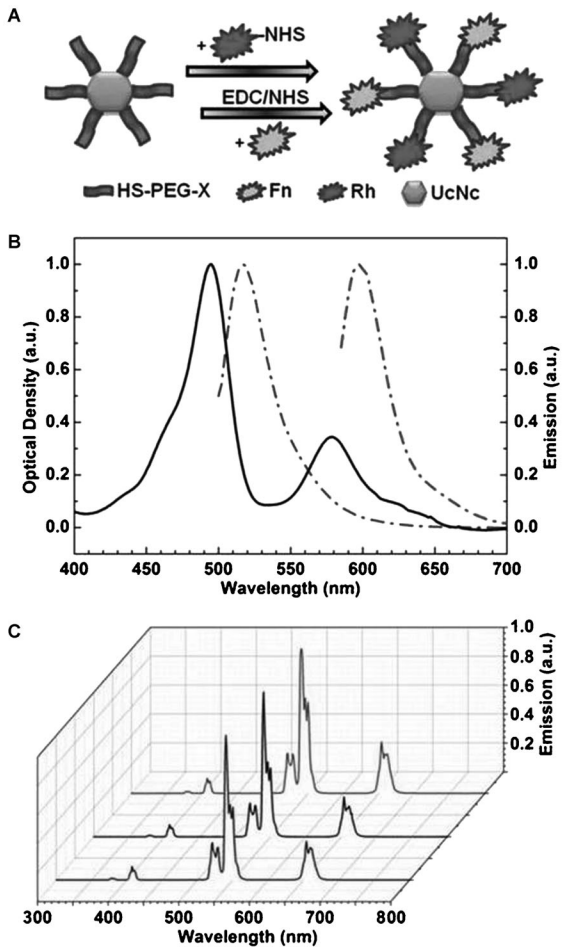

Figure 4. A) Schematic illustration of the reaction process on UCNCPcn. Usually, as a first step, the amines on the surface of UCNCs are covalently linked to NHS-modified dyes (such as Rh-NHS) through an amide bond. Then, the superficial carboxylic acids are linked to amine dyes (such as Fn) through EDC/NHS chemistry. B) Absorption spectrum of the dual-functional system (UCNCPcnRhFn, solid line) and fluorescence spectra (dash-dotted line) of the covalently linked dyes (left: fluorescein; right: rhodamine 101). All spectra are normalised to the maximum, and fluorescence spectra were collected by irradiation at the maximum absorption wavelength of the dyes. C) Top to bottom: fluorescence spectra obtained after irradiation (CW 980 nm laser at 189 mW) of UCNCPcn after reaction with i) fluorescein, ii) fluorescein and rhodamine 101, and iii) rhodamine 101. Fluorescence spectra were normalised at 540 nm.

ure S6, Supporting Information). This behaviour is associated with the fact that the PEG ligands do not interfere with the emission mechanism, nor do they completely shield the UCNC surface from solvents.[2a] Digital pictures of the total upconversion fluorescence of a milliQ solution of UCNCPcn under 980 nm irradiation or white light are shown in Figure 3 (see Figure S4, Supporting Information, for UCNCPcn in acetonitrile). The upconversion efficiency of the nanocrystals decreased when they were placed in water, but they exhibited a similar intensity to that of oleate-capped NPs in organic solvents (Figure S4, Supporting Information), confirming that the decrease in luminescence was due solely to the solvent molecules and not to the ligand exchange process. In fact, the presence of water quenches the UCNC emission by increasing the multiphonon relaxation of the lanthanide excited state owing to the high-energy vibrational modes of water.[2a,27] On the other hand, despite the quenching effect of water, the emission signals of PEG-capped UCNCs were sufficient for biological applications in living cells.[28]

We obtained stable water-dispersible nanostructures with either amino or carboxylate surface groups or a combination of the two. This allowed their selective covalent conjugation to a wide variety of molecules (Figure 4 A). For testing of the reactivity of the amino and carboxylic groups on PEG-capped UCNCs, the nanocrystals were conjugated to fluorophores through amide bonds by using peptide chemistry (EDC/NHS, EDC = *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide, NHS = *N*-hydroxysuccinimide).[10,11] Control experiments were also performed by using similar processes but without the reactive molecules (i.e., EDC/NHS and NHS-modified molecules) in the mixture; this allowed us to rule out any non-specific adsorption on the surface of the UCNCs or within the coating. The products were reacted

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and stored in triethanolamine (TEA) buffer (pH 7.5) to avoid competitive interaction between the UCNC and strong coordinating molecules contained in other buffers (such as phosphate (PO_4^{3-}) in PBS).[2a]

In addition, after several hours, no type of flocculation of the UCNCs in the medium for the cell experiments (DMEM) was detected.[2a] The abbreviations for the reactants used in this work are Fn for fluorescein cadaverine and Rh for rhodamine 101; the position of the abbreviations in the name of the samples indicates the reaction sequence, for example, UCNCPcnRhFn represents UCNCs coated with the 1:1 mixture of Pn (HS-PEG-NH_2) and Pc (HS-PEG-COOH) and conjugated with rhodamine 101 (Rh) and then with fluorescein cadaverine (Fn).

UV/Vis and fluorescence spectra of UCNCPcn derivatised with both Fn and Rh were obtained to confirm the success of the reactions (Figure 4B). The absorption bands of both dyes are present in the spectra, and the overlap between the emission of Fn and the absorption of Rh is clearly visible. Notably, the spectra of the control samples of UCNCPcn, prepared using only one of the reactive dyes and adding the other in an unreactive form, show the absorbance band of only the covalently linked dye (Supporting Information, Figure S7). This demonstrates that 1) the groups present on the surface of the nanocrystals are reactive, and 2) non-specific adsorption of the dyes on the coating is negligible.

Emission spectra of UCNCPcnRhFn, UCNCPcnFn, and UCNCPcnRh were recorded in TEA buffer and are reported in panel C of Figure 4 (see Figure S8 in Supporting Information for digital pictures). The shape of the emission spectra of the dye-modified UCNCs does not show significant changes, but the green signal of UCNCPcnFn seems to be decreased slightly. This is probably caused by an energy transfer from the green emission of UCNC and fluorescein. It could be interesting to investigate the energy transfer and interactions between UCNCs and their covalently linked payloads in order to develop novel synergistic smart nanoprobes.[9] It is worth noting that, assuming a steric occupancy[16,29] of 0.35 to 1 nm^2 (depending on the nanoparticle size) for every PEG chain on the UCNCs and a nearly quantitative yield of the EDC/NHS reaction, the estimated maximum number of dyes for each nanocrystal is 500 (25000 for 50 nm UCNCs; see Supporting Information). These values were useful for estimating the molar concentration of UCNCs in solution (on a nanomolar scale) if coupled with the quantification of dyes obtained through the ligand replacement.

We also successfully tested a protocol in which UCNCs were directly encapsulated by dye-modified Pn and Pc (data not shown). In this process, the polymer was first linked covalently to the dyes through peptide chemistry, and then, the product was grafted to the UCNCs by the free thiol. Thanks to this small variation, the control of the number of payload per particle is increased, but two possible drawbacks are under study: 1) not all payloads can be used in this process (e.g., proteins), and 2) the payload could affect the number of ligands per particle.

For the determination of the suitability of using PEG-capped UCNCs for biological and medical applications, the nanocrystals were tested and analysed in living cells, through both confocal microscopy and TEM. As a proof of concept, the experiments were carried out on the nanostructures of 10 nm in diameter.

The toxicity and intracellular uptake of all three types of PEG-encapsulated UCNCs (UCNCPn, UCNCPc, and UCNCPcn) was assessed by using both optical and TEM cell-imaging techniques in order to shed light on the dependence of the internalisation and toxicity of the particles on the charge and concentration of crystals in the medium. These assays were performed using differentiated mouse neural stem cells.

These cells typically give rise to the three main cell populations present in the central nervous system: neurons, astroglia, and oligodendrocytes.[30] Thus, this system better resembles the physiological conditions of the brain than those found in established cell lines. The cytotoxicity of UCNCs is a critical aspect when using them as bioprobes. Our results show that UCNCs have a low level of toxicity (Figures 5, and Figures S9, and S10 of the Supporting Information). This is in agreement with previous reports showing that rare-earth elements[31] and UCNCs[4,32] have good chemical stability and generally low toxicity. Four different concentrations (0, 1, 10, and 100 $\mu g\,mL^{-1}$) of UCNCPn, UCNCPc, and UCNCPcn were tested on the cells for 24 and 72 h. The results of the 24 h incubation with UCNCPn are shown in Figure 5 (see Figure S9 in the Supporting Information for UCNCPc and UCNCPcn). Control (0 $\mu g\,mL^{-1}$) cells and cells treated with 1 $\mu g\,mL^{-1}$ and 10 $\mu g\,mL^{-1}$ were apparently healthy and not stressed when photographed in culture (Figure 5, left column), although a 100 $\mu g\,mL^{-1}$ concentration resulted in cell death over vast areas. The effect of UCNCs on cell proliferation was assessed with SH-SY5Y cells by using a commercial toxicity kit based on the cleavage of yellow tetrazolium salt XTT, to form an orange product (formazan) by metabolically active cells (Figure 5M). The proliferation of the control cells was assumed to be 100%. After 24 h incubation with UCNCs with 1 and 10 $\mu g\,mL^{-1}$ concentrations, no effect on the cell viability was measurable. When the concentration of nanocrystals was increased to 100 $\mu g\,mL^{-1}$, cell proliferation dropped by less than 10%. Only with a 500 $\mu g\,mL^{-1}$ concentration was the toxicity of UCNCs evident, with a reduction in viable cells of 80%. These safe working conditions in the 1–100 $\mu g\,mL^{-1}$ range are in accordance with those published[33] by others for UCNCs and different cell types, including mouse mesenchymal stem cells.[34] Semi-thin sections showed no significant alterations in the cell membrane or in cytoplasmic and nuclear content at lower concentrations of UCNCs (Figure 5, middle column). However, at the highest 100 $\mu g\,mL^{-1}$ concentration, irregular membranes, cellular aggregates, condensed nuclear chromatin, and even picnotic cells were visible. This was especially evident in the case of UCNCPc-treated cells at this highest concentration, suggesting that optimal biological conditions are achieved with

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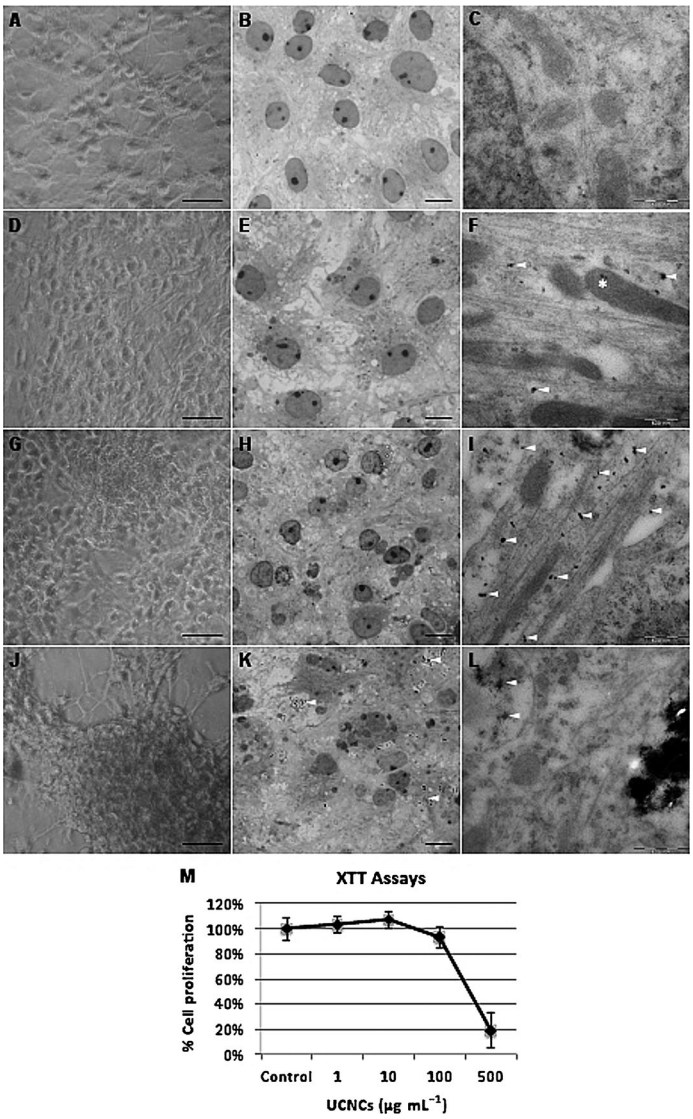

Figure 5. Viability study of internalised UCNCPns after 24 h incubation (left column: cell culture images; middle column: semi-thin sections; right column: ultra-thin TEM images). A)–C) Control cells (0 $\mu g\,mL^{-1}$). D)–F) 1 $\mu g\,mL^{-1}$-treated cells with no visible signs of stress. Nanocrystals are observed as electrodense particles (arrowheads, F), even in mitochondria (asterisk). G)–I) 10 $\mu g\,mL^{-1}$-treated cells contain a higher number of visible UCNCPns (arrowheads, I). J)–L) 100 $\mu g\,mL^{-1}$-treated cells. Cultures show signs of stress (J), cellular alterations such as irregular membranes and picnotic cells are noticeable (arrowheads, K), and UCNCs tend to form aggregates in the cells causing cell damage (arrowheads, L). M) In vitro cell proliferation of SH-SY5Y cells incubated with UCNCs at different concentrations for 24 h. Scale bars: left column, 200 μm; middle column, 10 μm; right column, 500 nm.

UCNCPn or UCNCPcn concentrations below 10 $\mu g\,mL^{-1}$. The intracellular uptake of UCNCs by these cells was confirmed by TEM (Figure 5, right column) and confocal microscopy experiments (Figure S11, Supporting Information). Nanocrystals were easily identified as electron-dense bodies and were abundant in the cytoplasm, but interestingly enough, they were also present in cellular organelles and in the nucleus, showing their ability to cross cell membranes. We assessed endocytosis as the most common uptake mechanism for UCNCs of similar sizes.[35] We were able to locate UCNCs in the cell membrane (Figure S12, Supporting Information), but none of the particles observed were associated with endocytic vesicles. However, classic endocytosis or clathrin-mediated endocytosis could not be discarded. The number of UCNCs was significantly higher in 10 $\mu g\,mL^{-1}$-UCNC-treated cells than in those with the lowest concentration. Nevertheless, in 100 $\mu g\,mL^{-1}$-treated cells, large aggregates were found and cellular damage was also observed. No differences were evident after 72 h of incubation of the cells with UCNCs except in the case of UCNCPcs, which started to become evident after 24 h of incubation, and thus, surprisingly, seems to be more harmful than UCNCPn or UCNCPcn at a 10 $\mu g\,mL^{-1}$ concentration (Figure S10 of the Supporting Information).

Two experiments were performed to assess whether the labelled nanocrystals were stable in the cellular environment. First, U2OS cells were incubated in presence of the dual-functional system (UCNCPcnRhFn), and the ability of the nanocrystals to enter inside cells with the two tags covalently linked was assessed through colocalisation experiments (Figure 6). Almost complete colocalisation of the fluorescence emission of the fluorophores was observed (Figure 6), as further confirmed by the excellent value of Pearson's correlation coefficient (0.556). Remarkably, this value suggests that leaching of the coating from the UCNCs is negligible, and, if present, the polymer coating alone is not internalised inside the cells.[36]

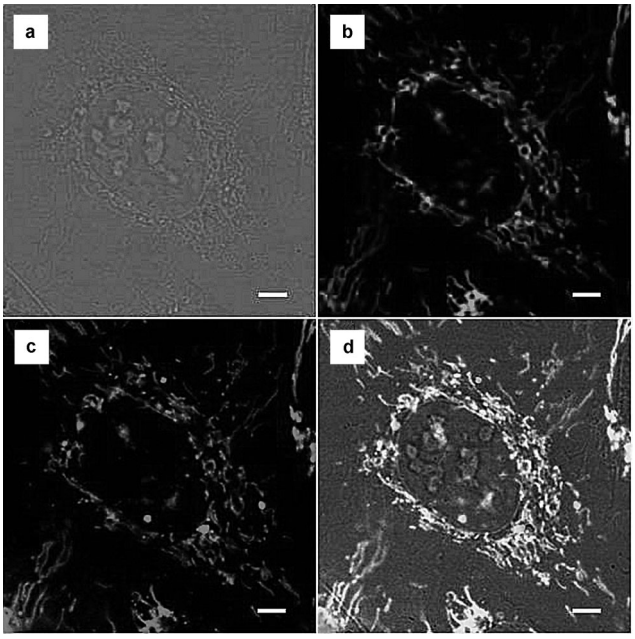

Figure 6. UcNcPcnRhFn administered to living osteosarcoma cells. a) Phase-contrast image; b) FITC fluorescence (exc.: 488 nm, em: 495–520 nm); c) RITC fluorescence (exc.: 561 nm, em: 595–620 nm); d) merging of a), b), and c). Pearson's correlation coefficient 0.556; scale bar 5 μm.

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In a further experiment, Förster resonance energy transfer (FRET) between the two fluorophores (Figure 4B) was considered as a marker of the structural integrity of the nanostructure. Upon irradiation at the excitation wavelength of the donor (Fn, 488 nm), strong FRET occurred between Fn and Rh, which caused nearly complete energy transfer to the red acceptor (Rh). As further confirmation of this hypothesis, the acceptor was irradiated at high laser powers, causing extensive photobleaching of the rhodamine dye. Thus, significant (2.2-fold) fluorescence enhancement of the donor signal was detected, confirming the FRET between the two dyes, and hence the structural integrity of the probe. These experiments were performed 2–4 h after the incubation of the nanostructures with living cells, suggesting excellent stability of the nanoprobes under physiological conditions.

Conclusion

A simple and quick strategy for the encapsulation of UCNCs synthesised in organic media has been devised to generate simple and modular systems that can be coupled easily to one or two different payloads through peptide chemistry. This process affords nanostructures that 1) are stable in water owing to the hydrophilicity of the PEGs used as the capping ligands, and 2) bear one or two orthogonal reactive groups (with either amino or carboxylic acid groups or both). One of the key points of this protocol is that the nanocrystal surface is not hidden (as occurs in other systems such as silica shells), so accurate investigation of their photochemical features in water solutions as well as their interactions with the medium (in particular, with covalently linked molecules) is possible. The success of the coating and the reactivity of the superficial groups to give rise to dual-functional nanocrystals were confirmed both in vitro and in living cells.

We also investigated the biocompatibility of the nanostructures by using cell-viability, confocal fluorescence, and TEM experiments; our protocol was tested with the ultimate goal of achieving targeted drug-delivery systems in the near future. We emphasise the versatility of our strategy and its direct applicability to the production of nanostructures of interest in biophysics and nanomedicine, such as building blocks of nanostructures with applications in sensing, imaging, and pharmacology.

Experimental Section

All chemicals used were purchased from Aldrich or Invitrogen unless otherwise noted, and were used as received. PEGs were purchased from Rapp-Polymer. The lanthanide chlorides were of 99.9% purity or higher.

Synthesis of upconverting $NaYF_4$:Yb^{3+} (18%):Er^{3+} (2.0%) nanoparticles: UCNCs were synthesised following a protocol reported elsewhere with some modifications.[13] Briefly, YCl_3 (0.8 mmol), $YbCl_3$ (0.18 mmol), and $ErCl_3$ (0.02 mmol) were mixed with oleic acid (12 mL) in a 50 mL round-bottomed Pyrex flask under continuous stirring. Octadecene (ODE, 15 mL) was added while the solution was heated to 160 °C. When everything had dissolved, the mixture was cooled to room temperature. A solution NaOH (2.5 mmol) and NH_4F (4.0 mmol) in methanol (10 mL) was added slowly into the flask. The solution was stirred while the solution was heated slowly to remove methanol, degassed at 100 °C for 10 min, heated to 300 or 315 °C, and finally, kept under Ar flux for one hour to yield 10 and 50 nm UCNCs, respectively. Then, the solution was cooled to room temperature and the nanocrystals were precipitated through the addition of ethanol. Subsequently, the oleate-capped UCNCs were washed three times with ethanol/water solution (1:1 v/v).

Encapsulation by mono-thiol hetero bifunctional PEGs: UCNCs (10 mg) were added to a degassed solution of HS-PEG-X 3 kDa [X = $-NH_2$ (Pn) or −COOH (Pc)] in $CHCl_3$ (2 mL). The solutions had either 50 mg mL^{-1} of one type of heterofunctional PEG or 25 mg mL^{-1} of Pn and 25 mg mL^{-1} of Pc for the 1:1 mix. Therefore, the conjugation process was performed in the presence of an excess (100 mg) of PEG.[2a] The mix was sonicated for 15 min, and then kept overnight in N_2 with strong stirring at 60 °C (Figure 1).

The opalescent solution was evaporated in vacuum and milliQ water (5 mL) was added. After sonication for 30 min, the colloid was filtered (0.22 μm filters) to remove large aggregates. Then, three cycles of centrifugation at 10000 rpm for 10 min with acetonitrile (a total of 60 mL) were performed to remove the oleate and the excess PEG. The PEG-encapsulated UCNCs (UCNCPc, UCNCPn, or UCNCPcn for the 1:1 mix) were resuspended in milliQ water (1 mL, or TEA, 1 mL, 40 mM, pH 7.5) and stored at +4 °C for a maximum of one week before use. Assuming about 0.6 nm^2 of surface covering for each PEG molecule,[16,29] the average number of PEG ligands per nanocrystal was 500 and 25000 for UCNCs with sizes of 10 and 50 nm, respectively; see Supporting Information.

Please note that the encapsulation process with Pn also worked at room temperature.

Reactions on superficial amines: A solution of UCNCPn or UCNCPcn (100 μL, 10 mg mL^{-1}), sonicated for 15 min, and 5(6)-carboxy-X-rhodamine *N*-succinimidyl ester (Rh-NHS, 0.5 mg) were added to TEA buffer (1 mL, 40 mM, pH 7.5). After stirring at RT for 5 h, the mixture was purified through five cycles of centrifugation/sonication in acetonitrile (centrifugation for 10 min at 10000 rpm; sonication for 15 min). The crystals (UCNCPnRh or UCNCPcnRh, see text for abbreviations) were resuspended in milliQ water or TEA (1 mL) and stored at +4 °C.

Reactions on superficial carboxylic acids: The following reagents were added in order to TEA (1 mL): 1) a solution of UCNCPc in milliQ water (100 μL, 10 mg mL^{-1}or UCNCPcn or UCNCPcnRh, see text for abbreviations), pre-sonicated for 15 min; 2) of fluorescein cadaverine (Fn) (100 μL, 20 mM solution in DMSO) as a proof of concept; and 3) EDC/NHS (100 μL, 21 mM). The mixture was stirred at RT for 5 h, and then purified through five cycles of centrifugation/sonication in acetonitrile (centrifugation, 10 min at 10000 rpm; sonication, 15 min). The crystals (UCNCPcFn or UCNCPcnRhFn) were resuspended in milliQ water or TEA (1 mL) and stored at +4 °C.

Dye quantification: We used a slightly modified method reported elsewhere[11] for the quantification of the number of fluorophores covalently linked to the encapsulated UCNCs. Briefly, 2-aminoethyl dihydrogen phosphate (15 mg) was added to a TEA solution (1 mL) of UCNCPcFn (or UCNCPcnRhFn or UCNCPnRh) (1 mg mL^{-1}) to release the dye-polymers from the surface of the UCNCs. Ligand exchange was allowed to proceed under stirring at RT overnight. Then, the solution was centrifuged (10000 rpm for 10 min) and the supernatant containing the free dye-polymer molecules was analysed by UV/Vis spectroscopy.

UV/Vis absorption measurements: All UV/Vis measurements were performed with an Agilent G1103A spectrometer equipped with quartz cuvettes. All samples were analysed in solutions of spectroscopic-grade solvents ($CHCl_3$, ACN, THF) or milliQ water and buffers. The concentrations of the solutions were adjusted to ensure that the maximum absorbance of each sample was below 0.1. All the reported spectra were normalised to the maximum absorbance.

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Fluorescence measurements: Fluorescence and upconverting emission measurements were performed with an SLM Amingo Bowmann series 2 fluorimeter. Standard procedures were used to collect fluorescence signals from dyes (solution in TEA at fixed pH of 7.5). The fluorescence signals from 0.5–10 $mg\,mL^{-1}$ solutions (organic solvents, milliQ water, and TEA buffer) of UCNCs were collected through irradiation of the samples with a CW 980 nm diode laser (Thorlabs L975P1WJ) coupled to the AB2 fluorimeter (Microbeam, S.A.).

FTIR measurements: The infrared measurements were performed on a Bruker Equinox 55/IR-Scope II equipped with a multiple reflection unit for attenuated total reflectance (ATR) measurements on solids. Usually, 1–2 drops of the UCNC solutions (0.5–2 $mg\,mL^{-1}$) were placed on the sample holder, and the spectra were collected after drying.

Transmission electron microscopy (TEM): TEM analyses were performed on a Jeol 1010 microscope (operating at 60 kV) equipped with a charge-coupled device (CCD) camera. Usually, for the preparation of UCNC samples, a drop of 1 $mg\,mL^{-1}$ solution of the colloids was left to dry under air at room temperature on a formvar/carbon film supported on a 300 mesh copper grid (3 mm in diameter).

Cells for TEM experiments were fixed in 3.5% glutaraldehyde for 1 h at 37°C. Cells were post-fixed in 2% OsO_4 for an additional 1 h at room temperature and stained in 2% uranyl acetate in the dark at 4°C for 2 h. Finally, cells were rinsed in distilled water, dehydrated in ethanol, and infiltrated in Durcupan resin overnight. Following the polymerization, the embedded cultures were detached from the chamber slide and glued to araldite blocks. Serial semi-thin sections (1.5 μm) were cut with an Ultracut UC-6 (Leica, Heidelberg, Germany), mounted onto slides, and stained with 1% toluidine blue. Selected semi-thin sections were glued with Super Glue-3 Loctite (Henkel, Düsseldorf, Germany) to araldite blocks and detached from the glass slide through repeated freezing (in liquid nitrogen) and thawing. Ultrathin sections (0.06–0.08 μm) were prepared with the Ultracut and stained with lead citrate. Finally, photomicrographs were obtained with a TEM FEI Tecnai G2 Spirit (FEI Europe, Eindhoven, Netherlands) with a Morada digital camera (Olympus Soft Image Solutions GmbH, Münster, Germany).

Dynamic light scattering: Measurements were performed in standard capillary cells DTS 1060 (for ζ-potential measurements) or a 700 μL standard cuvette (for size measurements) on a Zetasizer nano ZS DLS (Malvern Instrument) following the manufacturer's instructions, using solutions with a pH of 7.5 (TEA, 40 mM). A single scattering angle of 90°and UCNC concentrations between 1 and 0.1 $mg\,mL^{-1}$ were used for these measurements. Each value reported is the average of five consecutive measurements.

Cell culture and uptake of coated UCNCs: U2OS cells were cultured in DMEM medium supplemented with 10% foetal calf serum at 37°C in 5% CO_2. Cells were plated two days before imaging onto 30 mm glass-bottom dishes. Nanostructure uptake was achieved by incubating the cells with UCNCPcnRhFn (10 $\mu g\,mL^{-1}$) for 1 h at 37°C in 5% CO_2. U2OS cells were imaged after having been washed three times with PBS buffer and rinsed with fresh complete medium.

Mouse subventricular-zone-derived neural stem cells were isolated and cultured as described previously[30,37] for toxicity and internalisation assays. The maintenance and use of animals was carried out in accordance with the European Union guidelines (Directive 2010/63/EU) and all procedures were approved by the Animal Ethical Committee of the Universitat de València (A1330331766378). Cells were cultured for neurosphere formation in DMEM/F-12 medium containing 20 $ng\,mL^{-1}$ mouse recombinant receptor-grade epidermal growth factor (bEGF; Millipore, Billerica, MA, USA), 10 $ng\,mL^{-1}$ basic fibroblast growth factor (bFGF, Peprotech, Rocky Hill, NJ, USA), 1% N2, and 1% B27 (Gibco BRL/Life Technologies, Carlsbad, USA) at 37°C in 5% CO_2. Cells grew as free-floating neurospheres and were subcultured every seventh day in vitro through mechanical dissociation.

Toxicity assays: In vitro cytotoxicity was measured by using an XTT Cell Proliferation Kit II (Roche, Mannheim, Germany) on SH-SY5Y cells. Cells were seeded onto a 96-well plate at a density of 5×10^4 cells per well, under 100% humidity, and were cultured at 37°C and 5% CO_2 for 24 h; different concentrations of UCNCs (0, 1, 10, 100, and 500 $\mu g\,mL^{-1}$, diluted in culture medium) were then added to the wells. The cells were subsequently incubated at 37°C under 5% CO_2 for 24 h. Thereafter, XTT labelling mixture (50 μL; 0.3 $mg\,mL^{-1}$) was added to each well and the plate was incubated at 37°C under 5% CO_2 for an additional 4 h. Then, the absorbance at 450 nm was measured with a standard scanning multiwell spectrophotometer (Thermo Scientific, Massachusetts, USA). Five replicates of each experiment were performed. The relative cell viability (%) related to the control wells containing cell culture medium without nanocrystals was calculated by using the following formula: Cell viability (%) = (mean of A_{450} value of treatment group/mean of A_{450} value of control) × 100%.

Intracellular uptake: Mouse neurospheres were mechanically dissociated and 80000 cells/well were seeded in 4-well Lab-Tek poly-ornithine-coated chamber slides (Nalge Nunc International, Naperville, IL, USA). Cells were differentiated in the presence of 2% foetal bovine serum and in the absence of mitogens. Four days later, fresh medium containing different concentrations (0, 1, 10, and 100 $\mu g\,mL^{-1}$) of UCNCPc, or UCNCPcn, or UCNCPn were added to the cells. After 24 and 72 h, the cells were photographed with a Zeiss Axiovert 40 CFL inverted light microscope (Carl Zeiss AG, Oberkochen, Germany), and subsequently processed for TEM observation.

Colocalisation analysis of upconversion nanocrystals: Cell fluorescence was measured with a Leica TCS SP5 SMD inverted confocal microscope (Leica Microsystems AG) interfaced with an Ar laser for excitation at 488 nm and with a HeNe laser for excitation at 561 nm. Glass-bottom Petri dishes containing cells were mounted in a thermostated chamber at 37°C (Leica Microsystems) and viewed with a 40× 1.4 NA oil immersion objective (Leica Microsystems). The pinhole aperture was set to 1.0 Airy. The images were collected by using a low excitation power at the sample (10–20 μW) and monitoring the emission by means of an Acousto optical beam splitter (AOBS)-based built-in detector of the confocal microscope. Fluorescence was collected for 495–520 and 595–620 nm ranges for Fluorescein (Fn) and Rhodamine (Rh), respectively. All images were background-subtracted and analysed with ImageJ software. The colocalisation extent was estimated using Pearson's colocalisation coefficient provided by the "colocalisation finder" plug-in of ImageJ. Reported values (± standard deviation) are the average of three to eight measurements recorded on the same Willco dish.

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